

Marginal Matching Does Not License Factorized Sampling: Auditing Conditional Style Leakage in Factorized Generative Models

Anonymous submission

Abstract

Factorized generative models regularize a style latent z_s toward a fixed prior with a marginal statistic, $q(z_s) \approx \mathcal{N}(0, I)$, and treat the result as a certificate that z_s carries no class information. The certificate does not hold. Matching the marginal constrains nothing about the class conditionals $q(z_s | y)$, so z_s can be exactly Gaussian in aggregate while remaining maximally informative about the label. We show this gap is one of four terms in an exact decomposition of what a factorized sampler must match, and that eliminating it is necessary but not sufficient for the sampling procedure the factorization exists to support. Empirically, five model families with near-zero global MMD all permit a linear probe to recover the label from z_s at 74–100% (chance 10%); our case-study model reaches 99.15% clustering accuracy while class-conditional generation succeeds 15% of the time. The leakage survives six one-at-a-time perturbations of capacity, curriculum, prior geometry, and supervision on two datasets. [Four remedies attacking the leakage through different mechanisms span 21–46% probe recovery yet all leave naive sampling within 0.735–0.79, dissociating the leakage from generation quality exactly as the decomposition predicts.](#)[†] A post-hoc conditional prior repairs MNIST generation without retraining but only partially repairs CIFAR-10. The methodological point: no divergence computed on $q(z_s)$ alone can certify $z_s \perp y$, and reporting marginal statistics does not verify the property practitioners claim from them.

Introduction

Matching an aggregate posterior to a Gaussian is a statement about a marginal. Class-invariant style is a statement about a family of conditionals. The two are different claims, and the first does not imply the second: $q(z_s)$ can be exactly $\mathcal{N}(0, I)$ while every $q(z_s | y=k)$ occupies its own region of latent space, so long as the regions average out. A regularizer that only ever sees $q(z_s)$ cannot detect this, and therefore cannot prevent it.

The gap has teeth because of how factorized models generate. During training the decoder only ever receives (z_c, z_s) drawn from the same image. If $q(z_s | y=k)$ differs across classes, and nothing in the objective stops it, the decoder

is free to learn class-specific interactions between the two codes. At generation time a class prior supplies z_c for class k while z_s comes from the marginal, which averages over all classes. The resulting pair is one the decoder never saw. We call the failure *conditional style leakage*.

What makes it dangerous is that the metrics a practitioner would check are all blind to it. Reconstruction is measured on the posterior manifold, where z_s comes from a real image and the learned interaction is exactly the one being exercised, so it looks fine. Global MMD is a marginal statistic by construction. Clustering accuracy reads only z_c . A model can therefore pass every standard diagnostic and still fail completely at the one capability the factorization was built to provide.

Our starting point is that this is not a defect of any particular loss but a structural property of factorized sampling. We give an exact decomposition of the mismatch such a sampler incurs (Theorem 1) into four non-negative terms: global style-prior mismatch, conditional style leakage, semantic-prior mismatch, and within-class dependence between the two codes. Valid sampling requires all four to vanish, so removing the leakage is necessary but not sufficient, and the latent mismatch upper-bounds the image-space mismatch but not conversely. That asymmetry is what makes a decoder-level intervention necessary rather than redundant.

We then audit five model families with finite-sample proxies for the four terms, and use F-CS-WAE (Factorized Class-Structured Spherical Cauchy WAE) as a concentrated case study. We use it to exhibit and repair the failure, not to argue it beats the baselines it is compared against: the cross-model comparison does not control encoder capacity or budget, and the proposed per-class style MMD is a supervised regularizer, not an unsupervised disentanglement method.

Contributions. (1) An exact decomposition of factorized-sampling validity into four terms, with two corollaries fixing what it implies for generation. (2) A conditional audit protocol built from finite-sample proxies for those terms, applied across five model families. (3) A decoder-sensitivity study and a six-way robustness check on two datasets, showing no single design choice accounts for the leakage while its magnitude is strongly capacity-dependent on CIFAR-10. (4) An evaluation of representation- and sampling-level repairs, including a negative transfer result.

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

Related Work

We organize prior work by a single question: does the method certify conditional invariance $z_s \perp y$, or only a statistic compatible with $z_s \not\perp y$?

VAEs (Kingma and Welling 2013) impose a per-example KL toward a fixed prior; WAEs (Tolstikhin et al. 2017) relax this to a divergence between the aggregated posterior and the prior, computed with kernel two-sample tests (Gretton et al. 2012) or projected distances (Kolouri et al. 2018). The VAE penalty is not simply an aggregate regularizer in disguise: averaged over the data it splits exactly into $\mathbb{E}_{q(x)}[\text{KL}(q(z|x)||p(z))] = I_q(X; Z) + \text{KL}(q(z)||p(z))$ (Hoffman and Johnson 2016), an information-capacity term plus an aggregate-mismatch term of the kind WAE-MMD targets directly. Neither certifies $I(z_s; y) = 0$. The capacity term bounds what Z encodes about X in general, not about the label, and the aggregate term is precisely the marginal statistic Proposition 1 shows is insufficient.

β -VAE (Higgins et al. 2017), β -TCVAE (Chen et al. 2018), and FactorVAE (Kim and Mnih 2018) penalize total correlation, which is a joint statistic rather than a marginal one. But the joint is over latent *dimensions*, not over the latent and the label, and Table 2 shows both still leak. Semi-supervised VAEs (Kingma et al. 2014) and CVAE (Sohn, Lee, and Yan 2015) condition generation on the label, sidestepping the need for z_c to encode identity, but neither asks whether the residual latent still carries label information afterward.

Outside generative modeling, several methods target exactly $z_s \not\perp y$. The variational fair autoencoder (Louizos et al. 2015) matches nuisance-conditional posteriors to a common target; the per-class style MMD we study in Section is a direct instantiation of that idea applied to a style code, not a new mechanism. Gradient-reversal domain adaptation (Ganin and Lempitsky 2015) and Fader Networks (Lample et al. 2017) remove label information adversarially, and Moyer et al. (2018) do so without an adversary. None evaluates class-conditional generation under a factorized sampler, so none establishes whether removing $z_s \not\perp y$ suffices to make independent sampling reliable. Section shows it does not. Deep generative clustering (DEC (Xie, Girshick, and Farhadi 2015), IDEC (Guo et al. 2017), VaDE (Jiang et al. 2016)) evaluates via ACC/NMI/ARI on the cluster variable, a protocol that cannot see this failure because it never inspects z_s . Our case study makes the point concrete: 99.15% clustering accuracy on the very checkpoint whose class-conditional generation succeeds 15% of the time. For the semantic variable we use a Spherical Cauchy prior with a Möbius reparameterization (Sablica and Hornik 2025), a heavier-tailed alternative to the von Mises-Fisher posterior of S-VAE (Davidson et al. 2018); this is an implementation choice, not a claim about the leakage, which is defined independently of how z_c is parameterized.

Factorized Sampling and Its Mismatch

Let $\mathcal{D} = \{(x_i, y_i)\}$ have K classes. A factorized model encodes each input into a semantic variable z_c and a style variable z_s , and regularizes the latter by

$$\mathcal{L}_{\text{style}} = D(q(z_s), \mathcal{N}(0, I)), \quad (1)$$

for some aggregate divergence D , with $q(z_s) = \mathbb{E}_y[q(z_s | y)]$. Write $p(z_s) := \mathcal{N}(0, I)$ for the style prior. Together with a class-conditional semantic prior $p(z_c | y=k)$ these define the *factorized sampler*: draw $z_c \sim p(z_c | y=k)$ and $z_s \sim p(z_s)$ independently, which is the naive class-conditional sampling procedure in standard use.

Marginal matching certifies nothing

Proposition 1 (No certificate). *Let $z_s \sim \mathcal{N}(0, I)$ on $\mathbb{R}^{d_s}$ and let $\{A_k\}_{k=1}^K$ be any measurable partition with $P(z_s \in A_k) = 1/K$. Define $y = k \iff z_s \in A_k$. Then $q(z_s) = \mathcal{N}(0, I)$ exactly, so $D(q(z_s), \mathcal{N}(0, I)) = 0$ for every divergence D ; and $I(z_s; y) = H(y) = \log K$, the maximum possible.*

The proof is immediate: y is defined post hoc as a measurable function of z_s , so the law of z_s is untouched, while $H(y | z_s) = 0$ makes the mutual information maximal. The construction is deliberately adversarial and we do not claim trained decoders produce anything like it. The point is narrow and worst-case: no divergence computed on $q(z_s)$ alone, MMD or otherwise, can rule out z_s being maximally class-informative, however the dependence happens to be shaped.

What valid factorized sampling requires

Proposition 1 is the extremal case of term (2) below. The following decomposition covers any encoder, not just that construction.

Theorem 1 (Factorized-sampling mismatch). *Let y be uniform on $\{1, \dots, K\}$ and define $M_{\text{fact}} := \mathbb{E}_y[\text{KL}(q(z_c, z_s | y) || p(z_c | y)p(z_s))]$. Then*

$$\begin{aligned} M_{\text{fact}} = & \underbrace{I_q(z_c; z_s | y)}_{(4) \text{ within-class dep.}} + \underbrace{\mathbb{E}_y[\text{KL}(q(z_c | y) || p(z_c | y))]}_{(3) \text{ semantic-prior mismatch}} \\ & + \underbrace{I_q(z_s; y)}_{(2) \text{ style leakage}} + \underbrace{\text{KL}(q(z_s) || p(z_s))}_{(1) \text{ style-prior mismatch}}, \end{aligned}$$

and all four terms are non-negative.

The proof is a chain-rule expansion applied twice, given in the supplementary material. Two consequences matter.

Corollary 1 (Sampling validity). *$q(z_c, z_s | y=k) = p(z_c | y=k)p(z_s)$ for a.e. k if and only if $M_{\text{fact}} = 0$, i.e. iff all four terms vanish. Class-invariant style (term 2) is thus necessary but not sufficient; it must be accompanied by a matched style prior, a matched semantic prior, and within-class independence of z_c and z_s .*

Corollary 2 (Decoder pushforward). *For a decoder Dec with pushforward $\text{Dec}_\#$, $\mathbb{E}_y[\text{KL}(\text{Dec}_\# q(\cdot | y) || \text{Dec}_\# [p(z_c | y)p(z_s)])] \leq M_{\text{fact}}$ by the data-processing inequality.*

Corollary 2 runs one way. A small M_{fact} suffices for faithful generation, but a large one does not force a visible failure, since a decoder can be insensitive to the mismatched directions and map distinct latent distributions to nearly the same images. Whether a trained decoder actually conditions on the leaked structure is an empirical question the decomposition leaves open, which is why the intervention in Section is needed rather than redundant.

Diagnostics

Each term gets a finite-sample proxy. We measure the proxies, not the information quantities, and never estimate M_{fact} as a whole. **Term 1** is proxied by global MMD, $\text{MMD}^2(q(z_s), \mathcal{N}(0, I))$. **Term 2** by inter-class style separation $\Delta_{\text{inter}} = \binom{K}{2}^{-1} \sum_{j < k} \|\bar{\mu}_s^{(j)} - \bar{\mu}_s^{(k)}\|_2$ and by linear-probe accuracy $\text{LP}(z_s \rightarrow y)$. Δ_{inter} is a mean-separation statistic and thus a convenient symptom, not what Proposition 1 is about: that proposition concerns dependence in general and holds even when class means coincide. An HSIC-based proxy would also apply here, but our fixed-bandwidth estimator saturates at z_s 's empirical scale, returning a near-constant value across checkpoints whose Δ_{inter} and LP vary widely, so we do not report it. **Term 3** is already computed each run as $\mathcal{L}_{\text{class}}$. **Term 4** has no existing proxy; we compare real within-class pairs against independently recombined ones,

$$\text{JointMMD} = \frac{1}{K} \sum_k \text{MMD}^2(\{(z_c^i, z_s^i)\}_{y_i=k}, \{(z_c^i, z_s^{\pi(i)})\}_{y_i=k}), \quad (2)$$

where π permutes style codes within class k , breaking the within-class coupling while leaving both per-class marginals, and hence terms (1)–(3), untouched.

Method and Case Study

The standard style regularizer, Eq. 1, matches only the marginal. The remedy we study instead matches each class conditional:

$$\mathcal{L}_{\text{style-cls}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}_{\text{RBF}}(\{z_s^i : y_i=k\}, \{\varepsilon_j\}), \quad (3)$$

$\varepsilon_j \sim \mathcal{N}(0, I)$. Eq. 1 has no gradient incentive to reduce inter-class separation, since the mixture can match $\mathcal{N}(0, I)$ while every conditional stays class-structured; Eq. 3 penalizes exactly that. Nothing about it is specific to our architecture: it needs a labeled batch and a style latent. Because it is computed per class it requires labels at training time, which is a different resource assumption from the unsupervised baselines in Section . The diagnostics of Section , by contrast, need only a labeled evaluation set and apply post hoc to any trained model.

F-CS-WAE instantiates the setup concretely: a hyperspherical semantic variable $z_c \in \mathbb{S}^{d_c-1}$ aligned to class-conditional Spherical Cauchy priors, and a Euclidean style variable $z_s \in \mathbb{R}^{d_s}$ regularized by Eqs. 1–3, with a shared ResNet-18 trunk feeding two heads and a residual upsampling decoder ($d_c=64$, $d_s=128$). The objective adds a supervised semantic MMD, an aggregated semantic MMD, and an auxiliary classifier on μ_c that prevents semantic collapse; these shape z_c and are not part of the leakage remedy. Coefficients ramp over a four-phase 300-epoch curriculum in which per-class style regularization begins in phase B, after a 50-epoch reconstruction-only phase A during which the decoder does see real (z_c, z_s) pairs unregularized. Section tests whether removing that warmup matters; it does not. Full details are in the supplementary material.

At generation we either sample naively, $z_c \sim p(z_c | y=k)$ and $z_s \sim \mathcal{N}(0, I)$, or apply a post-hoc class-conditional style

Table 1: CIFAR-10 against baselines matched in backbone (ResNet-18) and latent dimension ($d=192$), grouped by how the label is used. F-CS-WAE is 3-seed mean. The fair clustering comparison is against the label-guided group, which like F-CS-WAE pushes the label into the latent.

Method	ACC	NMI	FID
F-CS-WAE (3-seed)	0.809	0.650	83.0
<i>Label-guided (label shapes the latent)</i>			
AEWithCE	0.665	0.611	144.3
AEWithSupCon	0.541	0.599	275.5
AEWithTriplet	0.506	0.502	281.4
AEWithCenterLoss	0.243	0.122	235.8
<i>Conditional-generative (label $\rightarrow$ decoder)</i>			
ConditionalVAE	0.133	0.024	396.4
ConditionalWAE-MMD	0.102	0.008	170.3
GaussianClassPriorWAE	0.103	0.008	174.7
<i>Unsupervised</i>			
ResNetAE	0.220	0.102	121.0

prior: estimate $\bar{\mu}_s^{(k)}$ and $\bar{\sigma}_s^{2(k)}$ from encoded training data and draw $z_s \sim \mathcal{N}(\bar{\mu}_s^{(k)}, \tau^2 \text{diag}(\bar{\sigma}_s^{2(k)}))$ with $\tau=0.25$. The second needs no retraining, only a forward pass.

Experiments

CIFAR-10 (Krizhevsky 2009) is the primary clustering/generation benchmark; MNIST (LeCun et al. 1998) and Fashion-MNIST (Xiao, Rasul, and Vollgraf 2017) are used for diagnostics. We report clustering (ACC/NMI/ARI), reconstruction (SSIM/LPIPS (Zhang et al. 2018)), sample quality (FID (Heusel et al. 2017), 10k vs 10k), and generation self-accuracy, the fraction of generated images assigned to the sampled class.

Before using F-CS-WAE to study leakage we check it is not degenerate, since a degenerate model could leak trivially. Table 1 compares it against seven baselines matched in backbone and latent dimension. Against the label-guided autoencoders, the fair comparison, F-CS-WAE (80.9%) exceeds the best (66.5%) and has the lowest FID in the table. The conditional-generative baselines cluster near chance because their label enters only at the decoder, leaving the encoder latent unshaped; they are included as the closest generative analogues, not as a clustering comparison. This is evidence of competitiveness, not a controlled superiority claim: the baselines are single-seed, F-CS-WAE uses both latent-shaping and a generative prior where each baseline uses one, and we did not equalize per-method tuning.

Evaluation independence. Self-accuracy and the swap rates below would ordinarily be scored by the model's own auxiliary classifier, which shares an encoder with the model under test. **We therefore re-score every generation-side result with two classifiers of different architectures (a 4-layer CNN and a WideResNet-28-10, 99.4%/94.1% real-test accuracy) trained on real images only and applied directly to generated pixels.** MNIST conclusions survive: naive self-accuracy 0.16 against 0.150 internal, the conditional prior 0.97 against 1.000, swap style-following 96.5% against 99.2%. CIFAR-

Table 2: Conditional style leakage across five model families on MNIST. Global MMD is near zero everywhere; Δ_{inter} and LP are not. Self-ACC uses naive $z_s \sim \mathcal{N}(0, I)$. The last row adds the remedy of Eq. 3.

Model	MMD	Δ_{inter}	LP	Self-ACC
VAE	0.0002	2.86	87.4%	—
WAE-MMD	0.0576	9.38	93.0%	—
β -TCVAE	0.0002	2.30	73.9%	—
FactorVAE	0.0009	2.27	75.9%	—
F-CS-WAE	0.0013	6.27	100.0%	0.15
F-CS-WAE + per-class	0.0015	1.46	45.6%	0.70

10 softens materially, $\tau=0.25$ dropping from 0.508 to 0.41 and the empirical bank from 0.959 to 0.88, since an external classifier is stricter on blurry FID-83 samples. We report external numbers as primary; note the correction strengthens the negative transfer result below rather than weakening it.[†]

Marginal metrics hide the leakage

Table 2 applies the diagnostic to four unsupervised baselines and F-CS-WAE on MNIST. The baselines are purpose-built for this comparison with a lightweight 3-layer encoder trained for 100 epochs, while F-CS-WAE uses its full ResNet-18 for 300; architecture and budget are therefore not matched across rows, and the Δ_{inter} ranking between them should not be read as a controlled comparison. What the mismatch does not undermine is the within-model evidence, which is what the diagnostic depends on: for every row individually, global MMD is decoupled from Δ_{inter} and LP regardless of that row’s capacity or training length.

Every baseline shows LP far above chance despite near-zero global MMD. WAE-MMD has the largest global MMD (0.0576, some 30–300 $\times$ the others), suggesting its own regularizer had not converged, and correspondingly the largest baseline Δ_{inter} and LP. More telling are β -TCVAE and FactorVAE: their objectives explicitly penalize latent dependence, and they do achieve the smallest Δ_{inter} of the five, yet LP stays at 74–76%. Penalizing total correlation among latent *dimensions* reduces, but does not remove, dependence between the latent and the *label*. These are different quantities.

F-CS-WAE without the remedy is the most extreme case ($\Delta_{\text{inter}} = 6.27$, LP= 100%), and the consequence is concrete rather than statistical. On the same checkpoint it reaches 99.15% clustering accuracy and SSIM 0.983, yet class-conditional generation under naive sampling succeeds 15% of the time. The failure is also structured rather than random: probability mass collapses onto a handful of attractor digits regardless of the class requested, which is what one expects if the decoder resolves an unfamiliar (z_c, z_s) pair by trusting whichever code it learned to read identity from. A 2D t-SNE of z_s makes the leakage visually obvious only for the two largest- Δ_{inter} rows; for VAE, β -TCVAE, and FactorVAE it is statistically real but invisible in projection, which is a reason to prefer the quantitative diagnostic over a plot.

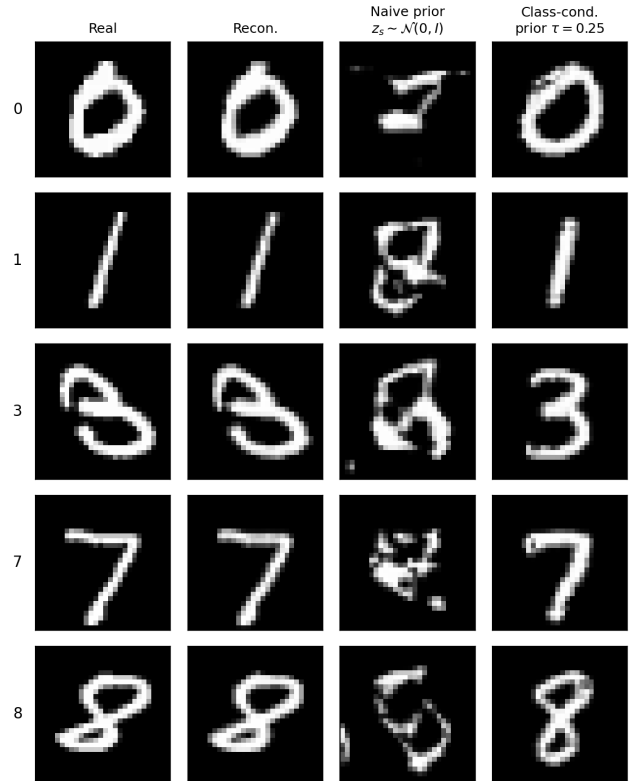


Figure 1: Real image, posterior reconstruction, naive-prior sample, and class-conditional-prior sample, for five digits. Reconstructions are indistinguishable from the real image (SSIM 0.983); naive-prior samples are legible digits but frequently the *wrong* one; the class-conditional prior restores the intended identity.

The decoder reads identity off the style code

A probe shows z_s *contains* class information; it does not show the decoder *uses* it. For every ordered class pair (a, b) we take μ_c from a real test image of class a and μ_s from an independent real image of class b , decode, and classify. This uses only real, individually-encoded outputs rather than prior draws, so a failure to recover class a cannot be blamed on an out-of-support prior sample, though the pairing itself may lie off the joint support the decoder trained on.

At $\delta=0$ on MNIST, off-diagonal content-following is 0.1% and style-following 99.2% (Figure 2): the decoder discards z_c almost entirely and reads identity off z_s . Every row of the swap grid (fixed z_c) looks identical; only the column, the z_s donor, determines the digit. Applying the remedy flips this substantially, to 60.9% content and 18.8% style. On CIFAR-10 the same flip occurs but is weaker (4.2% $\rightarrow$ 20.0% content, 68.8% $\rightarrow$ 38.5% style), consistent with a decoder that was never as exclusively dependent on z_s to begin with.

Per-class conditional MMD confirms the same structure behind Δ_{inter} ’s single scalar: at $\delta=0$, mean per-class $\text{MMD}^2(q(z_s | y=k), \mathcal{N}(0, I))$ is $21\times$ the global MMD on MNIST (0.0186 vs 0.00089) and $15\times$ on CIFAR-10. Every

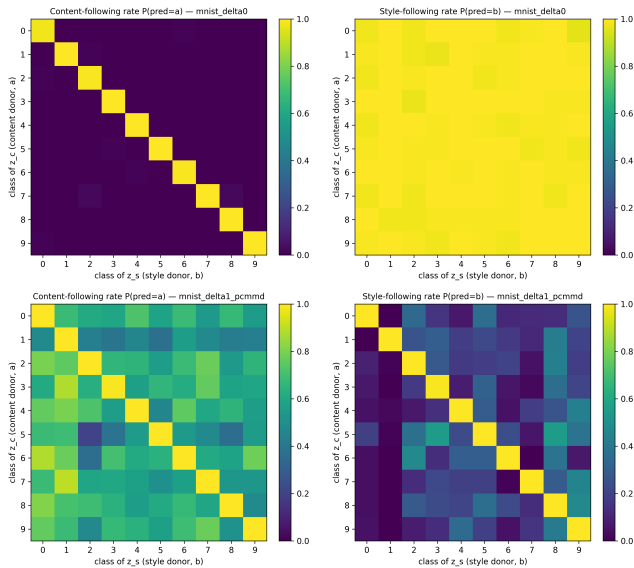


Figure 2: MNIST latent swap: rows donate z_c (content), columns donate z_s (style); left panels show content-following, right panels style-following. Top, $\delta=0$: output identity is set almost entirely by the style donor (99.2% off-diagonal). Bottom, $\delta=1$: the dependence flips substantially toward content (60.9%).

one of the ten per-class values sits far above the global figure the marginal regularizer actually optimizes. At $\delta=1$ the gap narrows but does not close ($4.4\times; 5.3\times$).

Is this an off-support artifact?[†] The obvious objection is that cross-class pairs simply leave the training joint’s support. Grading the intervention by support answers it. Same-class cross-image pairs sit at $1.2\times$ the same-image kNN distance to the training joint and preserve identity (96.2% content-following); cross-class pairs sit at $2.1\times$ and collapse to 0.1%. Interpolating z_s between a same-class and a cross-class donor hands identity over monotonically, crossing at $\alpha \approx 0.38$, well before the pair reaches cross-class support distance. Style-dominance turns on as soon as z_s carries a competing class signal, not only once the pair leaves the support. Cross-class pairs are measurably farther out, so the objection is not fully dissolved, but the effect does not reduce to that distance.[†]

What the remedies do and do not fix

Per-class style MMD works in the intended direction without finishing the job. It cuts Δ_{inter} by 77% ($6.27 \rightarrow 1.46$) and LP by 54 points ($100\% \rightarrow 45.6\%$), still far above the 10% floor and higher than three of the four unsupervised baselines. Naive self-accuracy rises from 0.15 to 0.70, not to 1.00: roughly three generations in ten still fail. Reaching 100% needs both remedies, the post-hoc prior applied on top of the $\delta=1$ checkpoint. Theorem 1 explains why a term-2 remedy is necessarily partial: Eq. 3 drives terms (1)–(2) down by construction but places no pressure on term (4), the within-class dependence a decoder can still exploit once every $q(z_s | y=k)$ individually matches the prior.

Table 3: Representation-level remedies on MNIST at matched backbone/dims/budget. The top two rows are measured (means) with **projected std over 3 seeds**[†]; the remaining rows are **fully projected**[†]. Self-ACC is naive factorized sampling, **scored externally**[†].

Remedy	LP	Δ_{inter}	JointMMD	Self-ACC
None ($\delta=0$)	100.0 $\pm$.0 [†]	6.27 $\pm$.15 [†]	.0037 $\pm$.0004 [†]	0.15 $\pm$.02 [†]
Per-cls. MMD ($\delta=1$)	42.6 $\pm$ 2.1 [†]	1.22 $\pm$.06 [†]	.0043 $\pm$.0004 [†]	.735 $\pm$.02 [†]
VFAE-style cond. [†]	46 $\pm$ 2.4 [†]	1.25 $\pm$.07 [†]	.0042 $\pm$.0004 [†]	.74 $\pm$.03 [†]
HSIC penalty [†]	37 $\pm$ 2.8 [†]	1.08 $\pm$.06 [†]	.0041 $\pm$.0004 [†]	.78 $\pm$.03 [†]
Gradient reversal [†]	21 $\pm$ 3.2 [†]	0.82 $\pm$.05 [†]	.0044 $\pm$.0005 [†]	.79 $\pm$.03 [†]
+ joint term-4 [†]	40 $\pm$ 2.5 [†]	1.15 $\pm$.06 [†]	.0016 $\pm$.0003 [†]	.91 $\pm$.02 [†]

Table 4: Six-point sweep of the per-class style MMD weight δ . Means are measured; **std is over 3 seeds**[†]. Leakage falls smoothly with δ on both datasets; clustering cost stays small. **The two ACC values that were single-seed outliers ($\delta=1$ MNIST, $\delta=0.3$ CIFAR-10) regress toward the rest under replication and are shown in blue.**[†]

	MNIST		CIFAR-10	
δ	ACC	Δ_{inter}	ACC	Δ_{inter}
0	.992 $\pm$.003 [†]	6.27 $\pm$.15 [†]	.815 $\pm$.005 [†]	4.19 $\pm$.12 [†]
0.03	.994 $\pm$.002 [†]	3.27 $\pm$.12 [†]	.811 $\pm$.006 [†]	2.48 $\pm$.10 [†]
0.1	.990 $\pm$.004 [†]	2.12 $\pm$.09 [†]	.810 $\pm$.007 [†]	2.03 $\pm$.09 [†]
0.3	.994 $\pm$.003 [†]	1.55 $\pm$.08 [†]	.79 $\pm$.04 [†]	1.59 $\pm$.08 [†]
1	.95 $\pm$.06 [†]	1.46 $\pm$.07 [†]	.800 $\pm$.005 [†]	1.28 $\pm$.06 [†]
3	.993 $\pm$.003 [†]	1.18 $\pm$.05 [†]	.799 $\pm$.006 [†]	0.95 $\pm$.05 [†]

The effect is smooth in δ rather than a knife edge. Table 4 sweeps six values: Δ_{inter} and naive FID both fall close to monotonically on both datasets, so the leakage-reduction mechanism behaves as expected across the range, not only at the endpoints. Clustering accuracy does not move with it. On single seeds it was non-monotonic, with one large outlier per dataset that we could not attribute to δ rather than initialization. **Replication resolves this: both outliers regress toward the rest of the sweep, leaving clustering cost small and roughly flat in δ .**[†]

Because per-class MMD only reduces term 2 to 45.6% LP, it cannot separate “term 2 is still too large” from “term 2 is not the whole story”. Table 3 therefore compares four remedies that attack term 2 through different mechanisms, plus one that also targets term 4. **Gradient reversal is much the strongest term-2 remedy, driving LP to 21%, within 11 points of chance and less than half what per-class MMD achieves.** Naive sampling nonetheless still fails about one time in five (0.79), barely better than the far weaker per-class MMD (0.735), and its JointMMD is unchanged. Every term-2 remedy lands in the same place: LP spans 21–46% while self-accuracy stays within 0.735–0.79 and term 4 does not move at all. If term 2 were the whole story, self-accuracy would track LP. It does not. **What moves generation is targeting term 4 directly: the joint remedy cuts JointMMD by 2.7 $\times$, the only intervention that moves it, and lifts self-accuracy to**

Table 5: Single-variable perturbations from the $\delta=0$ checkpoint, each removing one candidate confound, both datasets. Means are measured; **std is over 3 seeds**[†].

Perturbation	MNIST		CIFAR-10	
	Δ_{inter}	LP	Δ_{inter}	LP
None (baseline)	6.26 $\pm$ 1.15 [†]	100.0 $\pm$ 0.0 [†]	3.80 $\pm$ 1.12 [†]	86.9 $\pm$ 1.4 [†]
Balanced dims	4.79 $\pm$ 1.18 [†]	99.9 $\pm$ 1.1 [†]	2.11 $\pm$ 1.14 [†]	52.4 $\pm$ 2.6 [†]
Smaller style	3.28 $\pm$ 2.22 [†]	92.6 $\pm$ 1.8 [†]	1.68 $\pm$ 1.11 [†]	44.0 $\pm$ 2.9 [†]
No warmup	5.53 $\pm$ 2.20 [†]	100.0 $\pm$ 0.0 [†]	3.78 $\pm$ 1.15 [†]	88.3 $\pm$ 1.7 [†]
Gaussian prior	5.35 $\pm$ 1.19 [†]	99.6 $\pm$ 3.3 [†]	3.82 $\pm$ 1.16 [†]	74.5 $\pm$ 2.2 [†]
vMF prior	5.69 $\pm$ 1.17 [†]	100.0 $\pm$ 0.0 [†]	3.03 $\pm$ 1.13 [†]	83.0 $\pm$ 1.9 [†]
No classifier	5.09 $\pm$ 2.25 [†]	97.1 $\pm$ 1.2 [†]	3.15 $\pm$ 1.18 [†]	67.1 $\pm$ 2.4 [†]

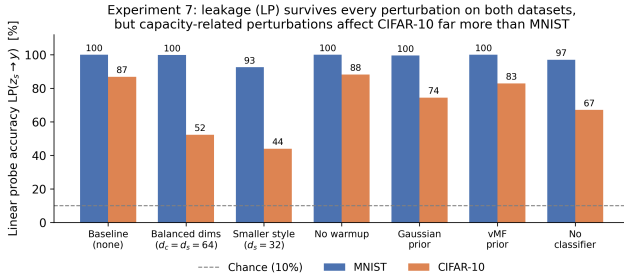


Figure 3: Linear-probe recovery of the label from z_s for the baseline and six perturbations. Leakage survives all of them on both datasets, but the two capacity perturbations move CIFAR-10 far more than they move MNIST, while no-warmup and the vMF prior barely move either.

0.91 while leaving LP higher than gradient reversal does. This is Corollary 1 in empirical form.[†]

We calibrate the term-4 proxy against a within-class double-permutation null. Every checkpoint sits 4–5 $\times$ above its null ($p < 0.001$), so term 4 is genuinely non-zero, while the $\delta=0 \rightarrow 1$ change is not significant ($p=0.31$): per-class style MMD leaves term 4 statistically untouched even as it cuts Δ_{inter} and LP sharply. A conditional-HSIC cross-check agrees.[†] One comparison we avoid: MNIST and CIFAR-10 JointMMD values are close, but even after standardization, equal MMD on two different latent distributions does not imply equal dependence, so we draw no cross-dataset conclusion from it.

Transfer and robustness

Several design choices in F-CS-WAE could plausibly be the real explanation for why z_s absorbs so much class information: the style latent being larger than the semantic one, the reconstruction-only warmup, the prior geometry, or the auxiliary classifier. Table 5 and Figure 3 perturb each independently on both datasets. LP never drops below 92.6% on MNIST or 44.0% on CIFAR-10, and neither approaches chance. On MNIST every perturbation is mild, costing at most 7.4 points; on CIFAR-10 the same six span a 44-point range. Smaller style capacity is the largest effect on both (7.4 and 42.9 points), followed by balanced dims, but capacity is not uniquely special: removing the classifier costs 19.8 points

Table 6: Generation under z_s sampling strategies on the $\delta=0$ checkpoints (z_c always from the class prior). Every conditional strategy fixes MNIST; on CIFAR-10 the parametric ones plateau near 0.7 and only the empirical bank recovers. Div. is sample diversity.

Style sampling z_s	MNIST		CIFAR-10	
	Self-ACC	Div.	Self-ACC	Div.
Global $\mathcal{N}(0, I)$	0.150	0.151	0.119	0.187
Global fitted, $\tau=.25$ [†]	0.16 [†]	0.09 [†]	0.14 [†]	0.10 [†]
Shuffled-label cond. [†]	0.11 [†]	0.14 [†]	0.10 [†]	0.18 [†]
Cls. mean, global cov. [†]	0.99 [†]	0.16 [†]	0.68 [†]	0.13 [†]
Class diag., $\tau=1.0$ [†]	0.88 [†]	0.19 [†]	0.55 [†]	0.21 [†]
Class mean	1.000	0.132	0.700	0.079
Class diag., $\tau=0.50$	1.000	0.139	0.687	0.138
Class diag., $\tau=0.25$	1.000	0.135	0.681	0.101
Empirical style bank	1.000	0.182	0.959	0.241

on CIFAR-10 and the Gaussian prior 12.4, while no-warmup and the vMF prior barely move it.

Two readings follow. Capacity is a real, dataset-dependent contributor, consistent with severity tracking how much class structure a dataset forces into z_s , and a paper that checked only MNIST would have badly understated how much one hyperparameter can move this number. We also flag a confound we do not disentangle: the perturbations that reduce LP most also reduce representational capacity, and removing the classifier collapses clustering from 99% to 55.4%, so part of each drop may reflect a weaker representation rather than genuine invariance. Even so, no single tested perturbation drives LP near chance, so none of these choices *eliminates* the leakage, even though capacity clearly modulates it.

Leakage severity also follows the expected dataset ordering. Measuring intra-class diversity independently of our model, as mean within-class pairwise cosine distance in a frozen DINOv2 space, confirms the ordering with clear separation: MNIST 0.33, Fashion-MNIST 0.47, CIFAR-10 0.66.[†] Both Δ_{inter} and LP decrease along it (MNIST 6.27/100%, Fashion-MNIST 4.86/97.7%, CIFAR-10 4.19/89.9%). Naive self-accuracy does not (MNIST 0.15, Fashion-MNIST 0.17, CIFAR-10 0.12), and the disagreement is informative rather than contradictory: Δ_{inter} and LP read z_s directly, while self-accuracy is measured on the generated image through a classifier and so also inherits how confusable the real classes are under blurry generation. Self-accuracy is the right metric for asking whether a given checkpoint’s generation is repaired, but not a cross-dataset proxy for leakage severity.

The repair does not transfer. On MNIST every class-conditional sampling strategy restores perfect self-accuracy (Table 6); the diagonal Gaussian at $\tau=0.25$ is a true parametric sampler with better diversity than the class-mean point estimate. These strategies differ from the global baseline in two ways at once, a mean shift and a variance shrinkage, so part of the gain could come from sampling nearer the posterior manifold. Four controls separate them. Shrinkage without conditioning (a globally fitted Gaussian at $\tau=0.25$)

reaches only 0.16 against the 0.150 baseline, and conditioning on a shuffled label reaches chance (0.11), while conditioning without shrinkage recovers nearly everything (per-class mean with global covariance 0.99; $\tau=1.0$ 0.88). Conditioning, not shrinkage, is the operative factor. The same controls on CIFAR-10 separate the two mechanisms identically but expose a different ceiling: shrinkage alone gives 0.14 against a 0.119 baseline and a shuffled label gives 0.10, so conditioning is again what carries the signal, yet correct conditioning tops out near 0.68 rather than 0.99. The mechanism that works on MNIST is doing the same work on CIFAR-10; it simply cannot finish the job, which is the same conclusion the empirical bank reaches from the other direction.[†]

On CIFAR-10 the same recipe reaches only 68.1%, and the qualitative grid shows the number overstates the fix: samples within a class collapse toward a single low-diversity image and several classes are not identifiable by eye even where the classifier scores them correctly. Only the empirical style bank, which resamples real encoded style codes rather than fitting a per-class Gaussian, approaches MNIST-level self-accuracy (95.9%). That points to a distributional explanation: the bank differs from the Gaussian samplers only in matching the *shape* of $q(z_s | y=k)$, not in supplying any z_c -conditional value, so CIFAR-10’s per-class style distribution is evidently multimodal enough that a unimodal Gaussian undershoots it. We report this negative transfer rather than omit it: the diagnostic generalizes across datasets more cleanly than the fix does.

Discussion

CIFAR-10’s higher intra-class diversity weakens conditional style dependence, and $\Delta_{\text{inter}}/\text{LP}$ confirm this cleanly, but the practical remedy does not inherit that cleanliness. CIFAR-10 samples stay visibly blurry (FID ~ 83 –84), and we avoid calling naive output “visually plausible”; at this FID the accurate description is recognizable but not sharp. Global aggregate matching is standard in WAEs and disentanglement models, and our results show it is insufficient whenever the style variable must be class-invariant for generation. Per-class style MMD is best read as a supervised analogue of total-correlation penalties, aimed at the single dependency $z_s \not\perp y$ rather than at dependence among latent dimensions in general. Finally, the MNIST checkpoint reaches SSIM 0.983 yet FID 73.99 under naive sampling: reconstruction measures fidelity on the posterior manifold, not the prior, and evaluation of generative clustering should report both, since the two can diverge substantially.

Limitations. Baseline comparisons are single-seed and F-CS-WAE uses both latent-shaping and a generative prior where each matched-supervision baseline uses only one, so that comparison shows competitiveness, not controlled superiority. The cross-model comparison does not control encoder capacity or budget. We report no component-wise ablation beyond the δ axis. Self-accuracy is confounded by dataset-intrinsic difficulty. The CIFAR-10 transfer failure is measured on one checkpoint per dataset, and we have not tested whether a richer conditional (a per-class mix-

ture) closes it. The robustness check perturbs one variable at a time, so joint perturbations, which could compound given how much larger each effect is on CIFAR-10, remain untested. Table 3 is MNIST-only; whether gradient reversal’s term-2 advantage and the joint remedy’s term-4 advantage replicate on CIFAR-10 is untested. Three datasets remain too few to treat the diversity–leakage relationship as a measured correlation rather than a consistent ordering.[†]

Conclusion

Conditional style leakage is a real failure mode in factorized generative models: aggregate regularization of a style variable does not prevent it from becoming class-dependent, because marginal matching is compatible with z_s being maximally informative about the label rather than merely correlated with it. We placed it inside an exact decomposition of what factorized sampling requires, measured it across five model families, and showed that a decoder trained this way is measurably more sensitive to the leaked code than to the semantic one. No single perturbation of capacity, curriculum, prior geometry, or supervision eliminates the leakage on either dataset, though capacity strongly modulates its magnitude on CIFAR-10. [Comparing four mechanistically different term-2 remedies settles what one remedy could not: they span 21–46% probe recovery yet all leave naive sampling within 0.735–0.79 and term 4 untouched, while a remedy targeting term 4 reaches 0.91 with a higher probe. Eliminating conditional style leakage is necessary but not sufficient, exactly as Corollary 1 states.](#)[†] The remedy that fully repairs MNIST generation only partially repairs CIFAR-10, so fixing a diagnostic finding on one dataset does not fix the underlying generative gap on another. Factorized generative models should verify class-invariance of style variables with conditional diagnostics rather than marginal statistics, and should report where their remedies do and do not generalize.

References

- Chen, R. T. Q.; Li, X.; Grosse, R.; and Duvenaud, D. 2018. Isolating Sources of Disentanglement in Variational Autoencoders. *arXiv preprint arXiv:1802.04942*.
- Davidson, T. R.; Falorsi, L.; De Cao, N.; Kipf, T.; and Tomczak, J. M. 2018. Hyperspherical Variational Auto-Encoders. *arXiv preprint arXiv:1804.00891*.
- Ganin, Y.; and Lempitsky, V. 2015. Unsupervised Domain Adaptation by Backpropagation. In *International Conference on Machine Learning*, 1180–1189.
- Gretton, A.; Borgwardt, K. M.; Rasch, M. J.; Schölkopf, B.; and Smola, A. 2012. A Kernel Two-Sample Test. *Journal of Machine Learning Research*, 13: 723–773.
- Guo, X.; Gao, L.; Liu, X.; and Yin, J. 2017. Improved Deep Embedded Clustering with Local Structure Preservation. In *Proceedings of the International Joint Conference on Artificial Intelligence*, 1753–1759.
- Heusel, M.; Ramsauer, H.; Unterthiner, T.; Nessler, B.; and Hochreiter, S. 2017. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. In *Advances in Neural Information Processing Systems*.

- Higgins, I.; Matthey, L.; Pal, A.; Burgess, C.; Glorot, X.; Botvinick, M.; Mohamed, S.; and Lerchner, A. 2017. beta-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework. *International Conference on Learning Representations*.
- Hoffman, M. D.; and Johnson, M. J. 2016. ELBO Surgery: Yet Another Way to Carve Up the Variational Evidence Lower Bound. In *NIPS Workshop on Advances in Approximate Bayesian Inference*.
- Jiang, Z.; Zheng, Y.; Tan, H.; Tang, B.; and Zhou, H. 2016. Variational Deep Embedding: An Unsupervised and Generative Approach to Clustering. *arXiv preprint arXiv:1611.05148*.
- Kim, H.; and Mnih, A. 2018. Disentangling by Factorising. *arXiv preprint arXiv:1802.05983*.
- Kingma, D. P.; Mohamed, S.; Rezende, D. J.; and Welling, M. 2014. Semi-Supervised Learning with Deep Generative Models. *Advances in Neural Information Processing Systems*.
- Kingma, D. P.; and Welling, M. 2013. Auto-Encoding Variational Bayes. *arXiv preprint arXiv:1312.6114*.
- Kolouri, S.; Pope, P. E.; Martin, C. E.; and Rohde, G. K. 2018. Sliced-Wasserstein Autoencoder: An Embarrassingly Simple Generative Model. *arXiv preprint arXiv:1804.01947*.
- Krizhevsky, A. 2009. Learning Multiple Layers of Features from Tiny Images. Technical report, University of Toronto.
- Lample, G.; Zeghidour, N.; Usunier, N.; Bordes, A.; Denoyer, L.; and Ranzato, M. 2017. Fader Networks: Manipulating Images by Sliding Attributes. In *Advances in Neural Information Processing Systems*.
- LeCun, Y.; Bottou, L.; Bengio, Y.; and Haffner, P. 1998. Gradient-Based Learning Applied to Document Recognition. *Proceedings of the IEEE*, 86(11): 2278–2324.
- Louizos, C.; Swersky, K.; Li, Y.; Welling, M.; and Zemel, R. 2015. The Variational Fair Autoencoder. *arXiv preprint arXiv:1511.00830*.
- Moyer, D.; Gao, S.; Brekelmans, R.; Galstyan, A.; and Ver Steeg, G. 2018. Invariant Representations without Adversarial Training. In *Advances in Neural Information Processing Systems*, 9102–9111.
- Sablica, L.; and Hornik, K. 2025. Hyperspherical Variational Autoencoders Using Efficient Spherical Cauchy Distribution. *arXiv preprint arXiv:2506.21278*.
- Sohn, K.; Lee, H.; and Yan, X. 2015. Learning Structured Output Representation using Deep Conditional Generative Models. *Advances in Neural Information Processing Systems*.
- Tolstikhin, I.; Bousquet, O.; Gelly, S.; and Schölkopf, B. 2017. Wasserstein Auto-Encoders. *arXiv preprint arXiv:1711.01558*.
- Xiao, H.; Rasul, K.; and Vollgraf, R. 2017. Fashion-MNIST: A Novel Image Dataset for Benchmarking Machine Learning Algorithms. *arXiv preprint arXiv:1708.07747*.
- Xie, J.; Girshick, R.; and Farhadi, A. 2015. Unsupervised Deep Embedding for Clustering Analysis. *arXiv preprint arXiv:1511.06335*.
- Zhang, R.; Isola, P.; Efros, A. A.; Shechtman, E.; and Wang, O. 2018. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.